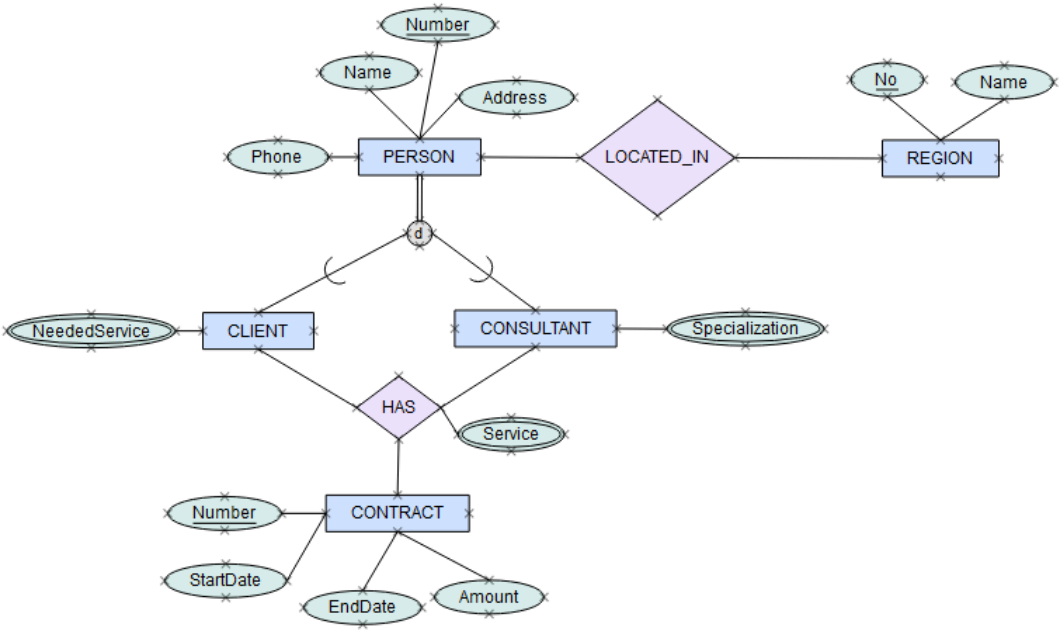
Status	Finished
Started	Sunday, 18 October 2026, 3:09 PM
Completed	Sunday, 18 October 2026, 5:09 PM
Duration	2 hours

Information

Information for questions 1-3:

The given EER diagram represents the database for maintaining information about clients, consultants and contracts. Each client/consultant lives in a single region. A client needs one or more services, and consultants have one or more services (specializations) they can provide. A contract usually involves a client and a contractor from the same region, although that is not always the case. Each contract needs to specify the services provided by the consultant to the client. A contractor may work on several contracts simultaneously, and there might be several contracts for the same client.

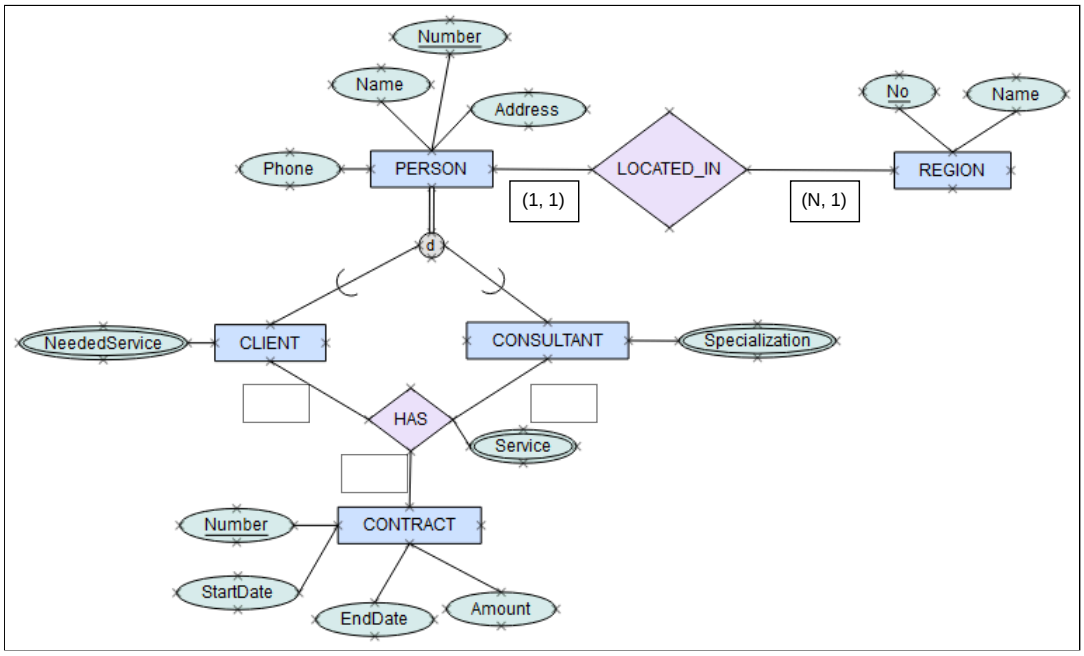


Question 1

Complete

Marked out of 5.00

[5 marks] Place the appropriate structural constraint onto each blank space of the EER diagram below, according to the specification given in the panel above.



(0, 0) (0, 1) (0, N) (1, 0) (1, 1) (1, N) (N, 0) (N, 1) (N, M)

Question 2

Complete

Marked out of 10.00

[10 marks] Map the completed EER diagram into a relational database schema. Specify the primary keys and any foreign keys.

Region(No, Name):

- Pkey: No

Person(Number, Name, Phone, Address, RegionNo)

- Pkey: Number
- Fkey: RegionNo references Region(No)

Client(Number, NeededService)

- Pkey: Number
- FKey: Number references PERSON(Number)

Consultant(Number, Specialization)

- Pkey: Number
- FKey: Number references PERSON(Number)

Contract (Number, StartDate, EndDate, Amount, ClientNumber, ConsultantNumber, Service)

- PKey: Number
- FKey: ClientNumber references CLIENT(Number)
- ConsultantNumber references CONSULTANT(Number)

Question 3

Not answered

Marked out of 10.00

[10 marks] The manager of the consulting company claims the database is not needed, as they can maintain all the necessary information in a sheet of an Excel file. They defined the following columns:

```
ClientNum, ClientName, ClientAddress, ClientPhone, ClientRegionNo,
ClientRegionName, ClientNeeds, ConsultantNo, ConsultantRegionNo,
ConsultantRegionName, ConsultantSpecializations, ContractNo,
ContractStartDate, ContractEndDate, ContractAmount, ContractServices.
```

The following columns may contain multiple values per cell: ClientNeeds, ConsultantSpecializations and Contract Services.

Discuss the potential problems with this Excel-based solution compared to using a relational database. Give examples of operations which might result in insertion, modification or deletion anomalies.

Information

Information for questions 4-6:

The following tables store data about employees, pilots and flights offered by a charter flight company. Pilots have two types of licences: ATP (Airline Transport Pilot) or COM (Commercial pilot). The licence needs to be renewed every 6 months. To fly commercially, pilots need to have a 2nd class medical certificate.

EMPLOYEE

Emp_No	FName	LName	Birthdate	Hire_Date
100	George	Bretton	15-Jun-62	15-Mar-08
101	Rhonda	Lewis	19-Mar-85	25-Apr-06
102	Rhett	Vandam	14-Nov-75	18-May-13
103	Robert	Williams	11-May-94	26-Jul-17
104	Anne	Johnson	12-Jul-91	20-Aug-10
105	John	Forster	14-Mar-95	19-Jun-17
106	Janine	Phillips	12-Feb-88	13-Mar-18
107	Paul	Smith	01-May-95	02-Jul-16
108	Robert	Williams	14-Jan-86	13-Jun-13
109	Elizabeth	Smith	19-Jun-81	14-Feb-16
110	Leigh	Black	31-May-63	29-Jun-10

PILOT

EmpNo	Licence	Licence_Date	Medical_Certificate
101	ATP	11-Jun-23	1
104	ATP	17-Jun-23	1
105	COM	15-Jul-23	2
106	COM	10-Aug-23	2
109	COM	12-Sep-23	1

FLIGHT

FlightNo	Date	Pilot	Copilot	HoursFlown	HoursWait
1001	10-Oct-23	104		5.1	2.2
1002	12-Oct-23	101		1.6	0
1003	15-Oct-23	105	109	7.8	0
1004	15-Oct-23	106		2.9	4.9
1005	16-Oct-23	109		5.7	3.5
1006	16-Oct-23	104	105	2.6	5.2
1007	16-Oct-23	106		7.9	0
1008	18-Oct-23	105		4.1	0
1009	19-Oct-23	109	105	8.6	22.5
1010	25-Oct-23	101	104	6.2	3.2
1011	25-Oct-23	101		1.9	5.3

Question 4

Complete

Marked out of 5.00

[5 marks] For each table, specify the primary/foreign keys it contains.

EMPLOYEE

primary key: [Your answer]

foreign key:

- -

PILOT

primary key: [Your answer]

foreign key:

- -

FLIGHT

primary key: [Your answer]

foreign key:

- -

Question 5

Complete

Marked out of 4.00

[4 marks] Specify two queries (the first one in relational algebra and the second one in SQL) to show all flights flown by employee 105 but not involving employee 109.

Relational Algebra

[Your answer]

SQL

[Your answer]

Question 6

Complete

Marked out of 6.00

[6 marks] For each of the following queries, state what the query does in natural language (English), and specify the result:

a. $\pi_{Pilot}(Flight)$

b. $\pi_{FName,LName,Hire_Date,Medical_Certificate,Licence_Date}((\sigma_{Flight=1003}(Flight)) \bowtie_{Copilot = EmpNo} Pilot) \bowtie_{Pilot.EmpNo = Employee.EmpNo} (EMPLOYEE)$

c. $Pilot\,\mathcal{F}_{COUNT\ FlightNo}(Flight)$

Part a

[Your answer]

Part b

[Your answer]

Part c

[Your answer]

Question 7

Complete

Marked out of 10.00

[10 marks] Define and contrast the following terms:

a. Key and superkey

b. Disjoint and overlapping specialization

c. Total and partial category

d. Conceptual and logical database design

e. Views and base tables

a. Key and superkey

[Your answer]

b. Disjoint and overlapping specialization

[Your answer]

c. Total and partial category

[Your answer]

d. Conceptual and logical database design

[Your answer]

e. Views and base tables

[Your answer]

Question 8

Not answered

Marked out of 10.00

[10 marks] What is the integrity component of the relational data model? Define each integrity constraint and discuss how this can be specified in SQL.

Question 9

Not answered

Marked out of 10.00

[10 marks] Relation R(A, B, C, D, E, G, H) is given, as well as a set of functional dependencies: F = {AB → C, DEG → H, A → B, DG → H}.

Find the minimal cover for F.

Question 10

Not answered

Marked out of 2.00

[2 marks] How many clustered indexes can be created on a table?

- Only one clustered index can be created on a table.
- There can be as many clustered indexes as there are fields in the table.
- One clustered index can be created for each non-key field in the table.

Question 11

Not answered

Marked out of 2.00

[2 marks] Refer to the following scenario:

- A table has 100 million records.
- Each row is 400 bytes.
- Each block is 8 kilobytes.
- Each index entry is 20 bytes.

> A binary search for a particular record reads approximately how many blocks?

- a. $\log_2(250{,}000)$
- b. $\log_2(5{,}000{,}000)$
- c. $\log_{10}(250{,}000)$

Question 12

Not answered

Marked out of 4.00

Select the correct type of index for each of the following situations:

1. Index with one entry for each table row:
2. Index with one entry for each table block:
3. An index on the table sort column:
4. An index that is not on the table sort column:

Question 13

Not answered

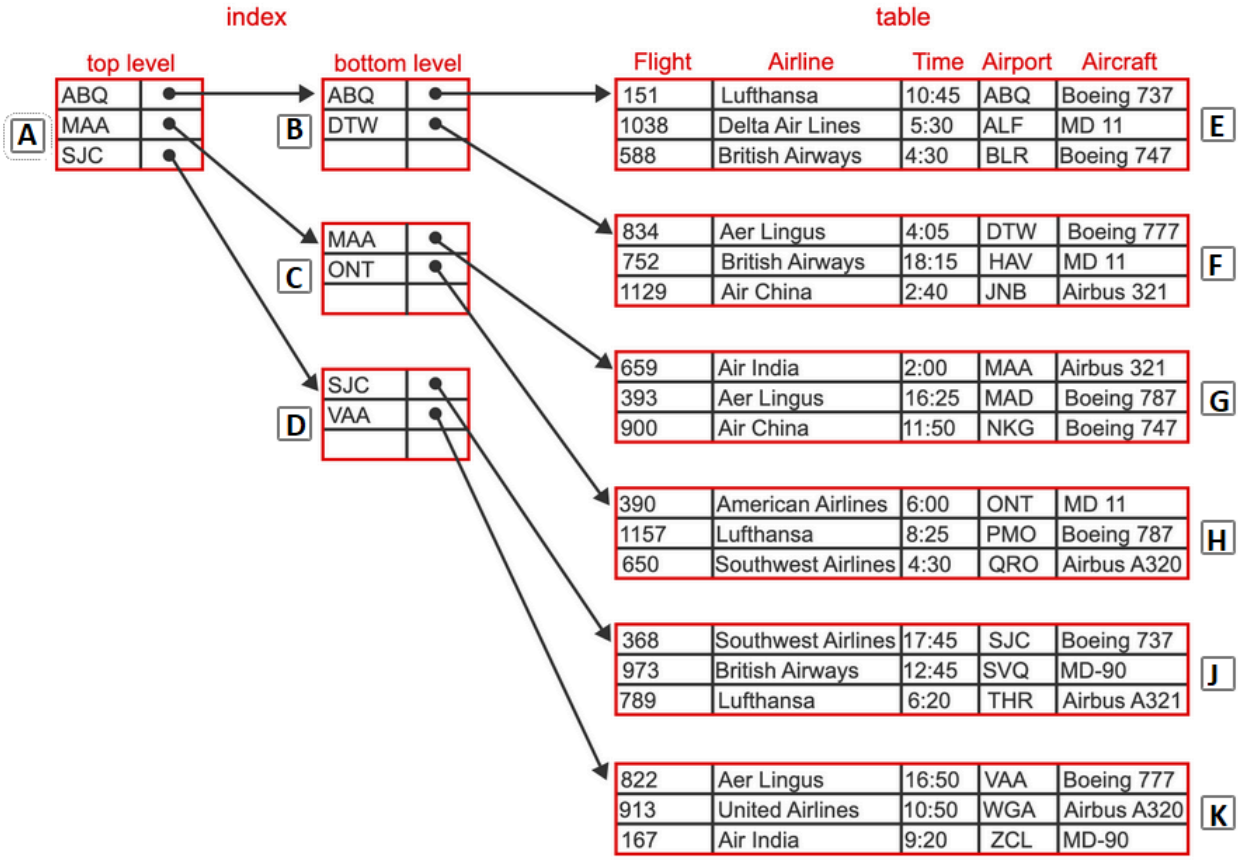
Marked out of 5.00

[5 marks] Assume the following:

- The ordering field is Airport.
- The index on Airport is sparse.

Indicate any blocks read when processing the following query:

```
SELECT Airline, Flight
FROM FlightTable
WHERE Airport LIKE 'WGA' OR Airport LIKE 'HAV';
```



- A
- B
- C
- D
- E
- F
- G
- H
- J
- K
- None of the above

Information

[9 marks this section] Select the correct type of conflict.

Question 14

Not answered

Marked out of 3.00

[3 marks] After T1 upgrades the seat, T2 determines the ticket cost. However, the upgrade is rolled back, so the customer is overcharged.

T_1	T_2
upgrade seat	
	read seat class
	write ticket cost
rollback	
	commit

- Dirty Read
- Phantom Read
- Nonrepeatable Read
- None of the above

Question 15

Not answered

Marked out of 3.00

[3 marks] T2 writes the ticket cost based on an economy seat. T1 later upgrades the seat, so the passenger gets first class priority. Cost and priority are inconsistent.

T_1	T_2
	read seat class
	write ticket cost
upgrade seat	
	read seat class
	assign boarding priority
commit	
	commit

- Dirty Read
- Phantom Read
- Nonrepeatable Read
- None of the above

Question 16

Not answered

Marked out of 3.00

[3 marks] T2 reserves a first class seat after T1 reads the first class seats, so T1 writes an incorrect seat total.

T_1	T_2
read first class seats	
	reserve first class seat
read economy seats	
write total seats	
commit	
	commit

- Dirty Read
- Phantom Read
- Nonrepeatable Read
- None of the above

Question 17

Not answered

Marked out of 2.00

[2 mark] Which of the following methods can be used for query optimization in a database?

- Indexing
- Sorting
- Hashing
- None of the above

Question 18

Not answered

Marked out of 1.00

[1 mark] Denormalization always improves the performance of the database.

- True
- False

Question 19

Not answered

Marked out of 5.00

[5 marks]

We would like to know the names of the students with a 3.5 GPA enrolled in 'COP 4720' during the Spring term 'Sp04'.

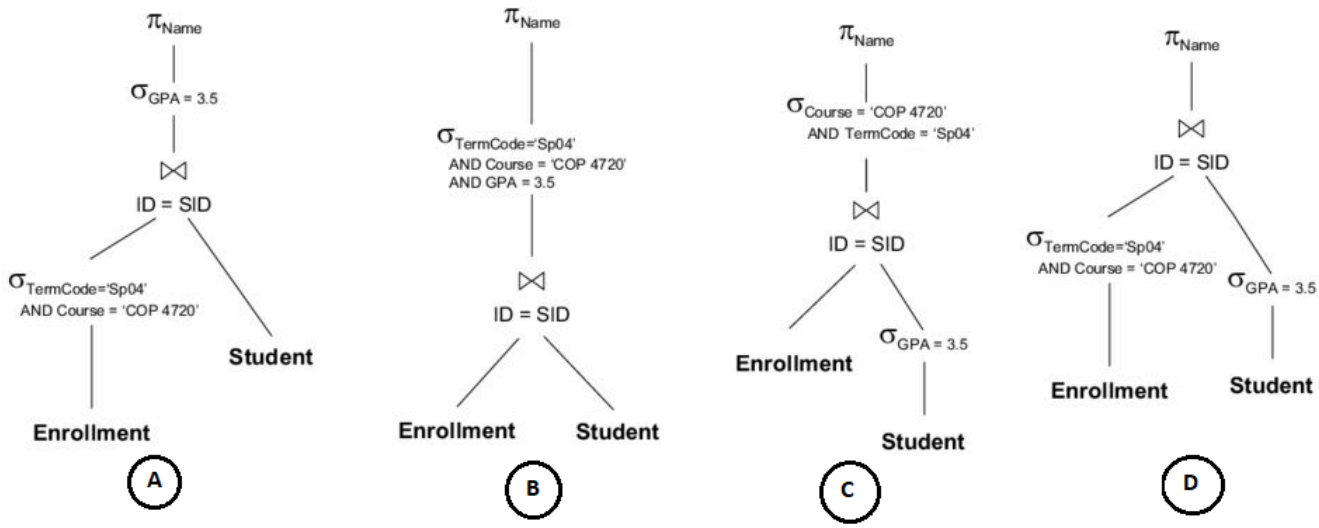
The information to answer this query is stored in this set of relations:

Student(SID, Name, DateOfBirth, GPA)

Enrollment(ID, Course, TermCode)

Course(Course, CourseName, Programme)

And the following query trees have been devised to answer this query:



Answer the following questions:

1. Which is the least efficient query tree? ____
2. Which is the most efficient query tree? ____
3. Assuming the following statement is executed successfully:
   CREATE INDEX my_index ON Student (Name);
   Would this affect the efficiency of the query trees? ____

Jump to...

2023 Exam: Solutions ►

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